Benchmarking Foundation Models using Satellite Image Time Series for Land Monitoring - MVA 2025/2026

Jean-François Guerrero jean-francois.guerrero@ens-paris-saclay.fr
Simon Sassi sassisimon84@gmail.com

April 2, 2026

1 Introduction

2 Dataset

We use the dataset PASTIS (Panoptic Agricultural Satellite TIme Series) There are 2433 sequences of multi-spectral of shape 10*128*128.Each sequences contains between 38 and 61 images. The satellite THEIA provides observations. The 10 channels correspond to the non-atmospheric spectral bands of the Sentinel-2 satellite, after atmospheric correction and re-sampling at a spatial resolution of 10 meters per pixel. Observatios are taken in several region of France with different climates, (Normandie, Bourgogne,Alsace, Provence). Almost 30 % of pictures are covered by cloud. [**dumeur**]

Each pixel is associated with a semantic label of 18 crop labels

3 Methods

4 Results

5 Summary of findings

Appendix A Method

Appendix B Data

Appendix C Tables

Table 1 Test metrics

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.5287	0.6526	0.8417
Tessera	0.5607	0.7018	0.8257
ALISE	0.6009	**0.7235**	0.8651
U-TAE	**0.7219**	-	**0.9170**

Table 2 Test metrics with scarce data (30 patches per fold)

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.2867	0.3755	0.7267
Tessera	0.4387	**0.5765**	0.7621
ALISE	**0.4482**	0.5729	**0.7898**
U-TAE	0.3869	0.5115	0.7043